INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL ACADEMIC LEARNING RESOURCE â¢ DEPARTMENT OF CSE

Course: [BCS-301] Data Structures

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: UNIT SUMMARY**

Module / Unit: Unit 3 **Faculty Author: Prof. S. Verma (Assistant Pro**

Unit 3 Quick Summary: Searching, Sorting & Asymptotic Complexities

==========

1. Searching Algorithms Comparison Summary

- Linear Search: O(n) worst-case time complexity, operates on unsorted arrays.
- Binary Search: O(log n) time complexity, strictly requires sorted random-access arrays.
- Hashing: Direct address tables, hash functions, collision resolution by chaining & probing.

2. Sorting Algorithms Master Complexity Table

- Bubble Sort / Insertion Sort / Selection Sort: Best O(n), Average O(n^2), Worst O(n^2).
- Quick Sort: Best O(n log n), Average O(n log n), Worst O(n^2) with poor pivot choice.
- Merge Sort: Guaranteed O(n log n) all cases; requires O(n) auxiliary memory space.

3. Graph Representation & Elementary Algorithms

- Representations: Adjacency Matrix (O(V^2) space) vs Adjacency List (O(V+E) space).
- Breadth-First Search (BFS): Queue-based level-order traversal, shortest paths in unweighted graphs.
- Depth-First Search (DFS): Stack/recursion-based traversal, topological sorting, cycle detection.